MI AED API

Version 1.1

REVISION HISTORY

Revision No.	Description	Date
1.0	● Initial release	09/12/2020
1.1	● Modified API	10/15/2020

TABLE OF CONTENTS

1. Overview

1.1. Module Description

AED (short for Acoustic event detection) is used to detect specific sound events in the audio stream. Currently supports baby crying and Loud Sound Detection (LSD).

1.2. Keyword

- Sensitivity

  Sensitivity of baby cry detection. The lower the sensitivity, the longer it takes for a continuous baby cry to be reported. On the contrary, the higher the sensitivity, the easier it is to be detected. The parameter range is listed according to AedSensitivity.

- Operating Point

  Baby crying detection operation point. Increasing the operating point will reduce the false alarm rate, and reducing the operating point will reduce the false alarm rate. This parameter needs to be in the range of [-10,10].

- Vad Threshold

  Voice Activity Detection dBFS threshold value. Must be higher than this setting value before entering baby cry detection.

- LSD Threshold

  Loud Sound Detection dBFS threshold. Above this set value, it is detected as loud sound.

1.3. Note

In order to facilitate debugging and confirm the effect of the algorithm, the user application needs to implement replacement algorithm parameter and grab audio data.

2. API REFERENCE

2.1. API List

API name	Features
IaaAed_GetBufferSize	Get the memory size required for Aed algorithm running
IaaAed_Init	Initialize the memory required for the Aed algorithm
IaaAed_Config	Configure Aed algorithm
IaaAed_Run	baby cry detection
IaaAed_GetResult	Get the result of baby cry detection

API name	**Features**
IaaAed_SetSensitivity	Set the sensitivity of baby cry detection
IaaAed_SetSampleRate	Set the data sampling rate for baby cry detection
IaaAed_SetOperatingPoint	Set the operation point for baby cry detection
IaaAed_SetVadThreshold	Set the Vad dBFS threshold of baby cry detection
IaaAed_SetLsdThreshold	Set the dBFS threshold for LSD
IaaAed_RunLsd	LSD algorithm processing
IaaAed_GetLsdResult	Get the results of LSD
IaaAed_Release	Quit Aed and release memory

2.2. IaaAed_GetBufferSize

➢ Features

Get the memory size required for Aed algorithm running .

➢ Syntax

```
unsigned int IaaAed_GetBufferSize(void);
```

➢ Parameters

Parameter name	Description	Input/output

➢ Return value

Return value is the memory size required for the operation of the Aed algorithm.

※ Dependency

- Header: AudioAedProcess.h
- Library: libAED_LINUX.so/ libAED_LINUX.a

※ Note

Only the required memory size is returned. The application and release of memory need to be processed by the application.

➢ Example

Please refer to IaaAed_Run example.

2.3. IaaAed_Init

➢ Features

Initialize the memory required for the Aed algorithm

➢ Syntax

AedHandle IaaAed_Init(char *working_buf_ptr, AedProcessStruct *aed_struct);

➢ Parameters

Parameter name	Description	Input/output
working_buf_ptr	Aed algorithm initialization parameter	Input
aed_struct	Aed initialization algorithm structure	Input

➢ Return value

Return value	Result
Not NULL	Successful
NULL	Failed

➢ Dependency

- Header: AudioAedProcess.h
- Library: libAED_LINUX.so/ libAED_LINUX.a

※ Note

- Whether using a baby cry detection or LSD function needs to call MI_AED_Init to initialize Aed algorithm to get the Aed algorithm handle.

➢ Example

Please refer to IaaAed_Run example.

2.4. IaaAed_Config

➢ Features

Configure Aed algorithm.

➢ Syntax

ALGO_AED_RET IaaAed_Config(AedHandle handle);

➢ Parameters

Parameter name	Description	Input/output
handle	Aed algorithm handle	Input

➢ Return value

Return value	Result
0	Successful
Non-zero	Failed

➢ Dependency

- Header: AudioAedProcess.h
- Library: libAED_LINUX.so/ libAED_LINUX.a

※ Note

None

➢ Example

Please refer to IaaAed_Run example.

2.5. IaaAed_Run

➢ Features

Aed algorithm baby cry detection processing

➢ Syntax

ALGO_AED_RET IaaAed_Run(AedHandle handle,short *audio_io);

➢ Parameters

Parameter name	Description	Input/output
handle	Aed algorithm handle	Input
audio_io	Baby cry detection data pointer	Input

- Return value

Return value	Result
0	Successful
Non-zero	Failed, refer to error code

- Dependency
  - Header: AudioAedProcess.h
  - Library: libAED_LINUX.so/ libAED_LINUX.a

※ Note

- The data length of audio_input must correspond to the point_number(the number of sampling points processed by the Aed algorithm at a time) set by AedProcessStruct.

- Example

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

#include "AudioAedProcess.h"

/*  0:Fixed input file  1:User input file   */
#define IN_PARAMETER 1

int main(int argc, char *argv[])
{
    AedHandle AED_HANDLE;
    AedProcessStruct aed_params;
    unsigned int aedBuffSize;
    char *working_buf_ptr = NULL;
    /***********************User change section start*************************/
    int vad_threshold_db = -40;
    int lsd_threshold_db = -15;
    int operating_point = -10;
    AedSensitivity sensitivity = AED_SEN_HIGH;
    AedSampleRate sample_rate = AED_SRATE_8K;
    aed_params.channel = 1;
    /***********************User change section end*************************/
    short input[1024];
    char src_file[128] = {0};
    FILE * fpIn;
    ALGO_AED_RET ret = ALGO_AED_RET_SUCCESS;
    int bcd_result;
    int lsd_result;
    int frm_cnt = 0;
    int lsd_db = 0; //useless param.

#if IN_PARAMETER
    if(argc < 2)
    {
        printf("Please enter the correct parameters!\n");
```

```
        return -1;
    }
    sscanf(argv[1], "%s", src_file);
#else
    sprintf(src_file, "%s", "./AFE_8K.wav");
#endif

    switch (sample_rate)
    {
        case AED_SRATE_8K:
            aed_params.point_number = 256;
            break;
        case AED_SRATE_16K:
            aed_params.point_number = 512;
            break;
        case AED_SRATE_32K:
            aed_params.point_number = 1024;
            break;
        case AED_SRATE_48K:
            aed_params.point_number = 1535;
            break;
        default:
            printf("Unsupported current sample rate !\n");
            return -1;
    }
    aedBuffSize = IaaAed_GetBufferSize();
    working_buf_ptr = (char *)malloc(aedBuffSize);
    if(NULL == working_buf_ptr)
    {
        printf("malloc workingBuffer failed !\n");
        return -1;
    }
    printf("malloc workingBuffer succeed !\n");
    AED_HANDLE = IaaAed_Init(working_buf_ptr, &aed_params);
    if(NULL == AED_HANDLE)
    {
        printf("IaaAed_Init faild !\n");
        return -1;
    }
    printf("IaaAed_Init succeed !\n");
    ret = IaaAed_Config(AED_HANDLE);
    if(ALGO_AED_RET_SUCCESS != ret)
    {
        printf("IaaAed_Config failed !, ret = %d\n", ret);
        return -1;
    }
    ret = IaaAed_SetSensitivity(AED_HANDLE, sensitivity);
    if(ALGO_AED_RET_SUCCESS != ret)
    {
        printf("IaaAed_SetSensitivity failed !, ret = %d\n", ret);
        return -1;
    }
    ret = IaaAed_SetSampleRate(AED_HANDLE, sample_rate);
    if(ALGO_AED_RET_SUCCESS != ret)
    {
        printf("IaaAed_SetSampleRate failed !, ret = %d\n", ret);
        return -1;
    }
    ret = IaaAed_SetOperatingPoint(AED_HANDLE, operating_point);
    if(ALGO_AED_RET_SUCCESS != ret)
    {
        printf("IaaAed_SetOperatingPoint failed !, ret = %d\n", ret);
```

```
          return -1;
        }
        ret = IaaAed_SetLsdThreshold(AED_HANDLE, lsd_threshold_db);
        if(ALGO_AED_RET_SUCCESS != ret)
        {
            printf("IaaAed_SetLsdThreshold failed !, ret = %d\n", ret);
            return -1;
        }
        ret = IaaAed_SetVadThreshold(AED_HANDLE, vad_threshold_db);
        if(ALGO_AED_RET_SUCCESS != ret)
        {
            printf("IaaAed_SetVadThreshold failed !, ret = %d\n", ret);
            return -1;
        }

        fpIn = fopen(src_file, "rb");
        if(NULL == fpIn)
        {
            printf("fopen in_file failed !\n");
            return -1;
        }
        printf("fopen in_file success !\n");
        memset(input, 0, sizeof(short) * 1024);
    #if 1
        /*Consider whether the input file has a header*/
        fread(input, sizeof(char), 44, fpIn); // Remove the 44 bytes header
    #endif
        while(fread(input, sizeof(short), aed_params.point_number * aed_params.channel, fpIn))

        {
            frm_cnt++;
            /* Run LSD process */
            ret = IaaAed_RunLsd(AED_HANDLE, input, &lsd_db);
            if (ALGO_AED_RET_SUCCESS != ret)
            {
                printf("MI_AED_RunLsd failed !, ret = %d\n", ret);
                break;
            }
            ret = IaaAed_GetLsdResult(AED_HANDLE, &lsd_result);
            if(ALGO_AED_RET_SUCCESS != ret)
            {
                printf("IaaAed_GetLsdResult failed !, ret = %d\n", ret);
            }
            if (lsd_result)
            {
                printf("current time = %f, loud sound detected!\n", \
                frm_cnt * ((float)aed_params.point_number / sample_rate));
            }
            /* Run AED process */
            ret = IaaAed_Run(AED_HANDLE, input);
            if(ALGO_AED_RET_SUCCESS != ret)
            {
                printf("MI_AED_Run failed !,ret = %d\n", ret);
                break;
            }
            ret = IaaAed_GetResult(AED_HANDLE, &bcd_result);
            if(ALGO_AED_RET_SUCCESS != ret)
            {
                printf("IaaAed_GetResult failed !, ret = %d\n", ret);
            }
            if(bcd_result)
            {
```

```
                printf("Baby cried at %.3f.\n", \
                frm_cnt * ((float)aed_params.point_number / sample_rate));
            }
        }
        printf("AED end !\n");
        ret = IaaAed_Release(AED_HANDLE);
        if(ALGO_AED_RET_SUCCESS != ret)
        {
            printf("IaaAed_Release failed !, ret = %d\n", ret);
        }
        fclose(fpIn);
        return 0;
    }
```

2.6. IaaAed_GetResult

- Features

    Get the result of baby cry detection

- Syntax

    ALGO_AED_RET IaaAed_GetResult(AedHandle handle, int *aed_result);

➢ Parameters

Parameter name	Description	Input/output
handle	Aed algorithm handle	Input
aed_result	Baby cry detection result	Output

➢ Return value

Return value	Result
0	Successful
Non-zero	Failed, refer to error code

➢ Dependency

- Header: AudioAedProcess.h
- Library: libAED_LINUX.so/ libAED_LINUX.a

※ Note

None

➢ Example

Please refer to IaaAed_Run example.

2.7. IaaAed_SetSensitivity

➢ Features

Set the sensitivity of baby cry detection

➢ Syntax

ALGO_AED_RET IaaAed_SetSensitivity(AedHandle handle,AedSensitivity sensitivity);

➢ Parameters

Parameter name	Description	Input/output
handle	Aed algorithm handle	Input
sensitivity	Baby cry detection sensitivity	Input

➢ Return value

Return value	Result
0	Successful
Non-zero	Failed, refer to error code

➢ Dependency

- Header: AudioAedProcess.h
- Library: libAED_LINUX.so/ libAED_LINUX.a

※ Note

None

➢ Example

Please refer to IaaAed_Run example.

2.8. IaaAed_SetSampleRate

➢ Features

Set the data sampling rate for baby cry detection

➢ Syntax

ALGO_AED_RET IaaAed_SetSampleRate(AedHandle handle,AedSampleRate srate);

➢ Parameters

Parameter name	Description	Input/output
handle	Aed algorithm handle	Input
srate	Sampling rate of baby cry detection data	Input

➢ Return value

Return value	Result
0	Successful
Non-zero	Failed, refer to error code

➢ Dependency

- Header: AudioAedProcess.h
- Library: libAED_LINUX.so/ libAED_LINUX.a

※ Note

None

➢ Example

Please refer to IaaAed_Run example.

2.9. IaaAed_SetOperatingPoint

➢ Features

Set the operation point for baby cry detection

➢ Syntax

ALGO_AED_RET IaaAed_SetOperatingPoint(AedHandle handle,int operating_point);

➢ Parameters

Parameter name	Description	Input/output
handle	Aed algorithm handle	Input
operating_point	Baby crying detection operation point	Input

➢ Return value

Return value	Result
0	Successful
Non-zero	Failed, refer to error code

➢ Dependency

- Header: AudioAedProcess.h
- Library: libAED_LINUX.so/ libAED_LINUX.a

※ Note

None

➢ Example

Please refer to IaaAed_Run example.

2.10. IaaAed_SetVadThreshold

➢ Features

Set the Vad dBFS threshold of baby cry detection, enter baby cry detection when it is higher than this threshold.

- Syntax

  ALGO_AED_RET IaaAed_SetVadThreshold(AedHandle handle,int threshold_db);

- Parameters

Parameter name	Description	Input/output
handle	Aed algorithm handle	Input
threshold_db	Vad dBFS threshold for baby crying detection	Input

- Return value

Return value	Result
0	Successful
Non-zero	Failed, refer to error code

- Dependency
  - Header: AudioAedProcess.h
  - Library: libAED_LINUX.so/ libAED_LINUX.a

※ Note

  None

- Example

  Please refer to IaaAed_Run example.

2.11. IaaAed_SetLsdThreshold

- Features

  Set the threshold of Aed algorithm loud sound detection

- Syntax

  ALGO_AED_RET IaaAed_SetLsdThreshold(AedHandle handle,int threshold_db);

- Parameters

Parameter name	Description	Input/output
handle	Aed algorithm handle	Input
threshold_db	LSD threshold, more than the specified dBFS it was considered loud sound.	Input

➢ Return value

Return value	Result
0	Successful
Non-zero	Failed, refer to error code

➢ Dependency

- Header: AudioAedProcess.h
- Library: libAED_LINUX.so/ libAED_LINUX.a

※ Note

None

➢ Example

Please refer to IaaAed_Run example.

2.12. IaaAed_RunLsd

➢ Features

Aed algorithm LSD processing

➢ Syntax

ALGO_AED_RET IaaAed_RunLsd(AedHandle handle,short *audio_input, int *lsd_db);

➢ Parameters

Parameter name	Description	Input/output
handle	Aed algorithm handle	Input
audio_input	LSD data pointer	Input
lsd_db	Current data decibels	Output

➢ Return value

Return value	Result
0	Successful
Non-zero	Failed, refer to error code

- ➢ Dependency
  - Header: AudioAedProcess.h
  - Library: libAED_LINUX.so/ libAED_LINUX.a

※ Note

- The data length of audio_input must correspond to the point_number(the number of sampling points processed by the Aed algorithm at a time) set by AedProcessStruct.

- ➢ Example

  Please refer to IaaAed_Run example.

2.13. IaaAed_GetLsdResult

- ➢ Features

  Get the LSD results of Aed algorithm

- ➢ Syntax

  ALGO_AED_RET IaaAed_GetLsdResult(AedHandle handle, int* lsd_result);

- ➢ Parameters

Parameter name	Description	Input/output
handle	Aed algorithm handle	Input
lsd_result	LSD results	Output

- ➢ Return value

Return value	Result
0	Successful
Non-zero	Failed, refer to error code

- ➢ Dependency
  - Header: AudioAedProcess.h

- Library: libAED_LINUX.so/ libAED_LINUX.a

※ Note

None

➢ Example

Please refer to IaaAed_Run example.

2.14. IaaAed_Release

➢ Features

Release Aed algorithm resource

➢ Syntax

ALGO_AED_RET IaaAed_Release(AedHandle handle);

➢ Parameters

Parameter name	Description	Input/output
handle	Aed algorithm handle	Input

➢ Return value

Return value	Result
0	Successful
Non-zero	Failed, refer to error code

➢ Dependency

- Header: AudioAedProcess.h
- Library: libAED_LINUX.so/ libAED_LINUX.a

※ Note

None

➢ Example

Please refer to IaaAed_Run example.

3. AED data type

3.1. AED data type list

Data type	Definition
AedProcessStruct	Aed algorithm initialization parameter structure type
AedHandle	Aed algorithm handle type
AedSensitivity	Aed algorithm baby cry detection sensitivity type
AedSampleRate	Aed algorithm baby cry detection data sampling rate

3.2. AedProcessStruct

- Description

  Define Aed algorithm initialization parameter structure type

- Definition

```
typedef struct {
        unsigned int point_number;
    unsigned int channel;
} AedProcessStruct;
```

- Member

Member name	Description
point_number	Sampling points, Give value according to sampling rate Sampling rate: sampling points 8K:256, 16K:512, 32K:1024, 48K:1536
channel	Audio data channels

※ Note

None

- Related data types and interfaces

  IaaAed_Init

3.3. AedHandle

- Description

Define the Aed algorithm handle type

➢ Definition

typedef void* AedHandle;

➢ Member

Member name	Description

※ Note

- None

➢ Related data types and interfaces

IaaAed_Init
IaaAed_Config
IaaAed_Run
IaaAed_GetResult
IaaAed_SetSensitivity
IaaAed_SetSampleRate
IaaAed_SetOperatingPoint
IaaAed_SetVadThreshold
IaaAed_SetLsdThreshold
IaaAed_RunLsd
IaaAed_GetLsdResult
IaaAed_Release

3.4. AedSensitivity

➢ Description

Define the sensitivity type of baby cry detection for Aed algorithm.

➢ Definition

```
typedef enum {
    AED_SEN_LOW,
    AED_SEN_MID,
    AED_SEN_HIGH
} AedSensitivity;
```

➢ Member

Member name	Description
AED_SEN_LOW	Low sensitivity
AED_SEN_MID	Medium sensitivity

AED_SEN_HIGH	High sensitivity

※ Note

None

➢ Related data types and interfaces

IaaAed_SetSensitivity

3.5. AedSampleRate

➢ Description

Define the sampling rate type of baby cry detection for Aed algorithm

➢ Definition

```
typedef enum {
    AED_SRATE_8K  = 8000,
    AED_SRATE_16K = 16000,
    AED_SRATE_32K = 32000,
    AED_SRATE_48K = 48000
} AedSampleRate;
```

➢ Member

Member 名称	Description
AED_SRATE_8K	Sampling Rate 8000Hz
AED_SRATE_16K	Sampling Rate 16000Hz
AED_SRATE_32K	Sampling Rate 32000Hz
AED_SRATE_48K	Sampling Rate 48000Hz

※ Note

None

➢ Related data types and interfaces

IaaAed_SetSampleRate

4. Error code

AED API error codes are shown as follow:

Table 4-1 AED API error code

Error code	Definition	Description
0x00000000	ALGO_AED_RET_SUCCESS	AED runs successfully
0x10000201	ALGO_AED_RET_INIT_ERROR	AED initialization error
0x10000202	ALGO_AED_RET_INVALID_CONFIG	AED Config is invalid
0x10000203	ALGO_AED_RET_INVALID_HANDLE	AED Handle is invalid
0x10000204	ALGO_AED_RET_INVALID_SAMPLERATE	AED sampling rate is invalid
0x10000205	ALGO_AED_RET_INVALID_POINTNUMBER	AED sampling points are invalid
0x10000206	ALGO_AED_RET_INVALID_CHANNEL	AED channel number is invalid
0x10000207	ALGO_AED_RET_INVALID_SENSITIVITY	AED detection sensitivity is invalid
0x10000208	ALGO_AED_RET_INVALID_CALLING	AED API call sequence error
0x10000209	ALGO_AED_RET_API_CONFLICT	Other APIs are running